CPS 2015

Roles and responsibilities

The Data Management Plan should clearly articulate how the PI and co-PIs plan to manage and disseminate data generated by the project. The plan should outline the rights and obligations of all parties as to their roles and responsibilities in the management and retention of research data, and consider changes that would occur should a PI or co-PI leave the institution or project. Any costs should be explained in the Budget Justification pages.

Data will be generated at each of our institutional sites. The PI at each institution will be responsible for following the data management guidelines described below, and for saving the data to the RIC servers. This process of transferring all data to a central location will be facilitated by our research project coordinator. Our data management guidelines will be reviewed at the kickoff meeting for the project, and revisited briefly at each of our quarterly meetings with the PIs.

Types of data

The Data Management Plan should describe the types of data, samples, physical collections, software, curriculum materials, or other materials to be produced in the course of the project. It should then describe the expected types of data to be retained and shared, and the plans for doing so. The DMP should cover how data are to be managed and maintained during the project.

We will manage all scientific and technical content generated from this proposal. Our products will include 3 broad types of data.

First, we will have data related to each of our proposed experiments. These will include the detailed protocols used for each experiment, the raw data collected from each experiment, and the experimental code used to collect those data. The relevant code will include both the data acquisition code, and the code for the controllers used in the experiment. Together, this information will be sufficient for recreating the circumstances of each data collection session.

Our second type of data will be the code and results for all analyses completed on experimental data. Specifically, we store all code necessary to recreate figures, tables, and other results reported in all published manuscripts.

The proposed foundational developments will produce substantial simulation code for controller development and testing. These and the outputs relevant to all key findings will also be managed.

Policies for access and sharing and appropriate protection and privacy

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The Data Management Plan should describe the period of time the data will be retained and shared; factors that limit the ability to manage and share data, e.g., legal and ethical restrictions on access to human subjects data; and provisions for appropriate protection of privacy, confidentiality, security, and intellectual property.

Data will be made available upon request, after the conclusion of each of study. The availability of data (raw and analyzed) will be advertised on the project's website. Access will be free, but the original data collector/creator/principal investigator retains the right to use the data for analysis before making it publicly available. No patient identifying information will be included in any of the available data files.

Select videos of the experiments will be distributed, with subject faces blurred so as to be unidentifiable. Experimental analyses results will be made available through peer-reviewed journal publications and conference presentations. The code needed to generate all published results will be documented clearly.

Software generated under the project will be released under the GNU Public License, and available for free download on the labs' websites. This includes code for each of our 3 experimental systems.

We commit to speaking with any qualified scientist who wishes to discuss our results or the technology employed to obtain our results.

There will be no copyright or licensing required for any of the data generated in this project.

Research records will be archived for a minimum of seven years after final reporting or publication of a project. If any questions regarding the research, including but not limited to research integrity allegations, are raised during the seven- year retention period, the PI will keep the records until such questions are fully resolved.

Data storage and preservation of access

The Data Management Plan should describe the mechanisms and formats for storing data and making them accessible to others, which may include third party facilities and repositories; and other types of information that would be maintained and shared regarding data, e.g. the means by which it was generated, detailed analytical and procedural

information required to reproduce experimental results, and other metadata.

The servers at the Rehabilitation Institute of Chicago will serve as the central repository for all collected data. Raw data files will be saved in either Matlab or comma-separated value (CSV) format. Code for converting between these formats will also be provided to ensure easy access. The contents of each data file, including the experimental conditions or the analysis procedures used during generation will be documented. All raw data will be de-identified to ensure subject anonynimity.

Upon request, data will be made available through the project's website, hosted on an archived server run by the RIC IT Team.

Additional possible data management requirements